

Energy Structure of the Parallel Key Geometric Flow and Its Comparison with the GENERIC and Onsager Frameworks

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Abstract

This paper analyzes the **energy structure** of the Parallel Key Geometric Flow (PKGF) and compares it with two major frameworks in non-equilibrium thermodynamics: the **GENERIC** (General Equation for Non-Equilibrium Reversible-Irreversible Coupling) formalism and **Onsagers linear response theory**.

PKGF is a linear operator flow that integrates a conservative commutator-type component with a dissipative component generated by an elliptic operator via complexification. Its energy decay structure exhibits certain similarities to the dissipative mechanisms in GENERIC and Onsager theory, while also differing in essential ways—particularly regarding the separation of energy and entropy, the role of degeneracy conditions, and the nature of the dynamical variables involved.

The aim of this paper is to provide a **careful and mathematically grounded comparison**, clarifying where PKGF aligns with existing non-equilibrium frameworks and where it diverges. No new results in GENERIC or Onsager theory are claimed.

Keywords: Parallel Key Geometric Flow, PKGF, GENERIC Framework, Onsager Reciprocity, Non-Equilibrium Thermodynamics, Energy Structure

1 Introduction

The Parallel Key Geometric Flow (PKGF) is defined by the evolution equation

$$\partial_t K = [\Omega, K] + \lambda \mathcal{D}(K),$$

where the conservative component $[\Omega, K]$ is of commutator type and the dissipative component $\mathcal{D}(K)$ is generated by an elliptic operator. These components are integrated through the complexification

$$\tilde{K} = K_{\text{core}} + iK_{\text{fluct}}.$$

A key feature of PKGF is that its dynamical variable K is not a physical state variable (such as density, momentum, or temperature), but rather an **operator describing the structure that governs the evolution of states**. This places PKGF at a different hierarchical level from GENERIC and Onsager theory, which operate on macroscopic state variables. Viewing intelligence as a physical process grounded in conservation laws and metriplectic flows is a perspective increasingly supported by contemporary physical theories of intelligence [Fagan 2026].

The purpose of this paper is to analyze the energy structure of PKGF and compare it with the GENERIC and Onsager frameworks, highlighting both structural similarities and essential differences.

2 Energy Structure of PKGF

The dissipative operator is defined by

$$\mathcal{D}(K) = -\nabla^* \nabla K - (VK + KV), \quad V(x) \geq cI > 0,$$

and the full evolution is given by

$$\partial_t K = [\Omega, K] + \lambda \mathcal{D}(K), \quad \lambda > 0.$$

2.1 Energy Functional

We consider the energy functional

$$E(K) = \frac{1}{2} \langle\langle K, K \rangle\rangle_{L^2}.$$

This quantity does **not** represent thermodynamic internal energy. Instead, it measures the **structural amplitude** (or L^2 -norm) of the operator K . Thus, energy decay in PKGF corresponds not to thermodynamic dissipation but to

> **simplification, abstraction, or forgetting of structural information** > (the D-phase in Noetics).

Since $\mathcal{D}$ is negative definite,

$$\frac{d}{dt} E(K(t)) = \langle\langle \mathcal{D}(K), K \rangle\rangle_{L^2} \leq 0,$$

and the commutator term does not contribute to the energy:

$$\langle\langle [\Omega, K], K \rangle\rangle = 0.$$

The mathematical consistency of such dissipative gradient flows and their structure-preserving numerical realizations is well-documented within the Onsager framework [Chen et al. 2024].

3 Comparison with the GENERIC Framework

GENERIC is formulated as [Gmela 2025]

$$\dot{x} = L(x) \nabla E(x) + M(x) \nabla S(x),$$

where

- L is a Poisson operator (antisymmetric),
- M is a dissipative operator (symmetric, positive semidefinite),
- and the **degeneracy conditions**

$$L \nabla S = 0, \quad M \nabla E = 0$$

ensure energy conservation and entropy production.

GENERIC	PKGF
Reversible term $L\nabla E$	Commutator term $[\Omega, K]$
Irreversible term $M\nabla S$	Elliptic dissipative term $\mathcal{D}(K)$
Energy decay	$\frac{d}{dt}E(K) \leq 0$

3.1 Structural Similarities

Both frameworks exhibit:

- a reversible–irreversible decomposition,
- a gradient-like dissipative structure,
- and a reversible component that preserves energy.

3.2 Essential Differences

3.2.1 (1) Hierarchical difference in dynamical variables

- GENERIC: macroscopic **state variables** x
- PKGF: **operators** K describing structural dynamics

Because PKGF operates at the operator level, **Lie algebraic commutators** arise naturally instead of Poisson structures.

3.2.2 (2) Poisson vs. Lie algebraic structure

GENERICs Poisson structure is rooted in symplectic geometry, whereas PKGFs commutator structure arises from operator algebra.

3.2.3 (3) Absence of energy–entropy separation

GENERIC separates E and S explicitly. PKGFs $E(K)$ is a structural norm and does not correspond to thermodynamic energy.

3.2.4 (4) Degeneracy conditions do not hold

GENERIC requires

$$M\nabla E = 0,$$

ensuring energy conservation by the dissipative term.

PKGF does **not** satisfy an analogous condition:

$$\mathcal{D}(K) \not\subset K.$$

This reflects a fundamental feature of PKGF:

> **structural dissipation directly reduces the structural amplitude**, > consistent with the D-phase in Noetics.

3.2.5 (5) Infinite-dimensional setting

PKGF is formulated on infinite-dimensional Hilbert bundles, while GENERIC is typically finite-dimensional.

4 Comparison with Onsager Theory

Onsager's linear response theory is given by

$$\dot{x} = -Lx,$$

where L is symmetric and positive definite. Modern unsupervised operator learning approaches for dissipative equations have also begun utilizing the Onsager principle [Chang et al. 2025].

4.1 Similarities

- PKGF's dissipative operator $\mathcal{D}$ is linear, analogous to Onsager's linear dissipation.
- Strong ellipticity yields regularization effects.

4.2 Differences

Onsager reciprocity (symmetry of the response matrix) generally does not hold for PKGF [Fuchs et al. 2018].

However, this is not a defect. In many intelligent or adaptive systems—such as control systems, learning systems, or circuits—**asymmetric feedback is intrinsic**, and PKGF's dissipative structure naturally reflects this.

5 Position of PKGF

From the above comparisons, PKGF is neither:

- an extension of GENERIC,
- nor a generalization of Onsager theory.

Instead, PKGF provides an **independent operator-theoretic framework** for modeling reversible–irreversible dynamics at the structural level.

Its energy structure aligns with certain aspects of GENERIC and Onsager theory, but differs fundamentally in:

- the nature of the dynamical variables,
- the absence of degeneracy conditions,
- and the operator-level interpretation of dissipation.

6 Conclusion

This paper has provided a mathematically grounded comparison between the energy structure of PKGF and the GENERIC and Onsager frameworks.

PKGF shares with these theories:

- a reversible–irreversible decomposition,
- energy decay induced by the dissipative component,
- and invariance of energy under the conservative component.

However, PKGF differs in essential ways:

- operator-level dynamical variables,
- Lie algebraic rather than Poisson structure,
- absence of energy–entropy separation,
- failure of degeneracy conditions,
- and infinite-dimensional analytic setting.

PKGF should therefore be viewed not as a replacement for existing non-equilibrium theories, but as an **independent operator-theoretic and geometric framework** for analyzing non-equilibrium dynamics at the structural level.

Future work includes the analysis of nonlinear PKGF, Galerkin approximations, and a more refined perturbation theory in infinite dimensions.

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